Ethan Deal

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EDUCATION

Duke Kunshan University | 昆山杜克大学 **May 2026 | Kunshan 昆山, CN**

Dual Degree, Computation and Design

- GPA: 3.6/4.0; 1560/1600 SAT; Awarded Merit Scholarship (60% Tuition)
- Data Structures, Algorithms, Discrete Math, Databases, Stats, Modeling and Predicting, Probability, Computer Organization

Duke University | B.S. in Computer Science **2026 | Durham, NC**

EXPERIENCE

BRIYA | Fullstack Developer **July - Aug 2025 | Washington, DC**

IT Intern **May - Aug 2024 | Washington, DC**

- Created a **MERN** stack classroom booking system supporting **5 campuses** and **100+ faculty**, including conflict-aware booking logic, drag-and-drop editing, and recurring events with multiple calendar views, fully responsive across desktop and mobile.
- Deployed backend infrastructure with REST APIs, Nginx, and SSL on Linode, achieving zero downtime since launch.
- Enrolled and configured **1000+** Chromebooks for student distribution, performing disk wipes and domain trust restorations.

DUKE UNIVERSITY | Data Automation Engineer **May - July 2025 | Durham, NC**

- Engineered a Python-based data processing pipeline to automate mailing list segmentation for **10,000+ user records.**
- Automated survey consolidation with Qualtrics workflows to centralize large-scale surveys into standardized spreadsheets.
- Developed PyQt desktop application to transform multi-event datasets into aggregated reports with venue-based filtering.

VINTORY | Graphic Designer and Prompt Engineer **May - Sep 2023 | Remote**

- Built a Midjourney prompt library to streamline generative design, trained ChatGPT to produce structured prompt arrays.
- Spearheaded the adoption of AI-based design automation. Created **30+** custom postcards and branded **100+** templates and landing pages for top vacation rental management firms, using Midjourney to generate stock photos.

PROJECTS

RAG Fact-Checking Extension **July 2025 - March 2026**

- Designed a Chrome extension in React with a Node.js/Express backend to verify highlighted claims in under **5 seconds**.
- Architected a **RAG** pipeline using **LangChain**, ChromaDB, and a locally hosted MiniLM embedding model.
- Implemented and benchmarked single-step agentic vs. deterministic two-step retrieval architectures, evaluating latency, retrieval accuracy, context leakage, and inducing misinformation through prompt injection.

FlightOps Database Design **Oct 2024 – Jan 2025**

- Designed, implemented, and scraped data for a relational database in MySQL, with **10+ interrelated tables** for flight data.
- Engineered a web scraping pipeline with Playwright to query real-time flight data, bypassing API and rate limitations.
- Developed backend logic with CRUD functionality to optimize flight paths, manage assignments, and update crew data.
- Compiled databases of over **10,000** airports, **7000** aircraft, and **600** airlines to support large-scale flight analysis.

SKILLS

Languages: Python, Java, R, SQL, HTML/CSS, JavaScript, C, MIPS Assembly
Frameworks & Tools: ReactJS, NodeJS, Docker, LangChain, MongoDB, ChromaDB, Linode
Design & Media: Adobe Creative Suite, WordPress, Maya, Unity

ADDITIONAL INFORMATION

Languages: English (native), Mandarin (fluent), Spanish (beginner)
Interests: Digital painting, travel photography, piano, guitar, ultimate frisbee, soccer